

Comprehensive Quantitative Trading Model Implementation Plan: Intraday & Swing

This document provides a unified, step-by-step implementation roadmap for both Intraday and Swing quantitative trading models. It integrates shared, evidence-based practices (such as Purged Cross-Validation and the Triple Barrier Method) while strictly differentiating the data horizons, feature decay, and purging scales required for each specific time horizon.

1. Shared Core Methodologies (Evidence-Based Foundations)

Both intraday and swing models must adhere to strict quantitative science standards to prevent lookahead bias and model overfitting, based on the academic frameworks established by researchers like Marcos López de Prado.

- **Purged Walk-Forward Cross-Validation:** Standard random splits (K-fold) cause severe data leakage in time-series analysis. Both models must split data chronologically and strictly purge any training data whose outcome bleeds into the validation set.
- **Triple Barrier Labeling:** Instead of predicting continuous raw returns, both models classify which barrier is hit first: Upper (Profit Take), Lower (Stop Loss), or Time (Maximum Holding Expiration).
- **Gradient Boosting & Tree Ensembles:** Algorithms like LightGBM, XGBoost, and Random Forests are preferred over linear models or basic neural networks due to their robustness to tabular data and ability to find non-linear feature interactions without immense computational overhead.
- **Cross-Sectional Normalization:** Raw features (like RSI, ATR, or Volume) must be transformed into Z-scores relative to the trading universe at that specific timestamp to isolate stock-specific alpha from broad market beta.

2. Intraday Model Implementation (Minute-by-Minute)

The intraday model is highly sensitive to market microstructure and requires high-frequency data engineering.

Component	Intraday Specification
Data Horizon	1 to 3 Years. Older data contains obsolete market microstructure, outdated HFT algorithmic behaviors, and irrelevant tick sizes.
Candle Structure	1-Minute, 5-Minute, or Volume Bars (Volume bars are highly recommended to statistically normalize massive volume spikes at the open/close).
Asset Universe	Top 50 to 100 most liquid stocks in the S&P 500 to ensure high-frequency execution is possible with minimal slippage.
News Catalyst Decay	Measured in minutes. A steep exponential decay factor is applied aggressively within the first 30 to 60 minutes of the release.
Purging & Embargoing	Purge by minutes (e.g., purge 60 minutes of training data if max holding time is 60 minutes). Embargo typically spans an overnight gap.
Overnight Gaps	Positions are closed before the end of the day. Rolling technical features must explicitly not cross overnight price gaps without adjustments.

3. Swing Trading Model Implementation (Multi-Day)

The swing model relies on deeper historical context to navigate various macroeconomic regimes and multi-day volatility trends.

Component	Swing Specification
Data Horizon	5 to 10 Years. Necessary to expose the model to different macroeconomic cycles (bull markets, bear markets, zero/high interest rates).
Candle Structure	1-Hour or Daily candles. Hourly candles are optimal for capturing finer entry momentum shifts than daily closes provide.
Asset Universe	S&P 500 constituents. Must strictly include delisted stocks (survivorship bias prevention) historically accurate to each year.
News Catalyst Decay	Measured in days. A major news event (like an earnings beat) will influence multi-day price action, requiring a much slower exponential decay factor.
Purging & Embargoing	Purge by days (e.g., if max holding time is 5 days, purge the last 5 days of training data before the validation set starts).
Targeting Constraints	Must enforce sector-neutrality constraints so the model does not over-allocate to a single trending sector (e.g., all capital into Tech).

4. Step-by-Step Execution Workflow

1. **Data Extraction & Alignment:** Pull historical candles (10yr for Swing, 3yr for Intraday) and align exact point-in-time news timestamps to the very next available trading candle to prevent lookahead bias.
2. **Feature Engineering:** Compute moving averages, ATR, and sentiment scores. Apply the appropriate time-horizon exponential decay to news.
3. **Cross-Sectional Scoring:** Transform all features into Z-scores relative to the trading

universe at that specific time to force the model to identify relative strength.

4. **Labeling Strategy:** Generate Triple Barrier targets (1 for profit take, -1 for stop loss, 0 for timeout) matched to the respective intraday or swing holding constraints.
5. **Splitting Architecture:** Apply Walk-Forward Purged Time-Series Split. Apply minute-based purging and embargoing for intraday, and day-based for swing.
6. **Model Training:** Train LightGBM or Random Forest on the Training set. Strictly use the Validation set for early stopping to prevent hyperparameter overfitting.
7. **Holdout Testing & Commission Modeling:** Evaluate the final model on the completely unseen Test set. Hardcode realistic slippage, bid-ask spread crossing, and broker commission costs into the final metrics to prove live-trading viability.